

26 Supersymmetric Field Theories

26.7 The Supercurrent

Like any other continuous global symmetry, supersymmetry leads to the existence of a conserved current [5]. The conservation and commutation properties of the supersymmetry current are operator equations that will remain valid even when supersymmetry is spontaneously broken, and that will therefore be useful to us in Chapter 29, when we consider theories of spontaneous supersymmetry breaking in a non-perturbative context. Also, the supersymmetry current is related to components of a superfield known as the *supercurrent* [6], that will be of fundamental importance in our treatment of supergravity in Chapter 31.

As we saw in Section 7.3, the existence of an ordinary global symmetry of the Lagrangian density under an infinitesimal transformation $\chi^\ell \rightarrow \chi^\ell + \epsilon \mathcal{F}^\ell$ (with χ^ℓ a generic canonical or auxiliary boson or fermion field and $\mathcal{F}^\ell$ a function of the canonical and auxiliary fields) leads to the existence of a current

$$J^\mu(x) \propto \sum_\ell \frac{\partial \mathcal{L}(x)}{\partial (\partial_\mu \chi^\ell(x))} \mathcal{F}^\ell(x),$$

which is conserved for fields satisfying the field equations, and generates the symmetry in the sense that the canonical commutation relations give

$$\left[\int d^3x J^0(x), \chi^\ell(y) \right] \propto \mathcal{F}^\ell(y).$$

The supersymmetry current requires a somewhat more complicated treatment for two reasons. One is that supersymmetry is only a symmetry of the action, not of the Lagrangian density or the Lagrangian. Instead, the variation of the Lagrangian density under an infinitesimal supersymmetry transformation is a spacetime derivative, which we may write in the form

$$\delta \mathcal{L} = \sum_\ell (\bar{\alpha} \partial_\mu K^\mu), \quad (26.7.1)$$

with K^μ a four-vector of Majorana spinors. In consequence, the supersymmetry current is not the usual Noether current. The Noether current is a four-vector N^μ of Majorana spinors defined by

$$\sum_\ell \frac{\partial_R \mathcal{L}}{\partial (\partial_\mu \chi^\ell)} \delta \chi^\ell \equiv -(\bar{\alpha} N^\mu), \quad (26.7.2)$$

whose divergence is given by the Euler–Lagrange equations as

$$\begin{aligned} (\bar{\alpha} \partial_\mu N^\mu) &= \sum_\ell \frac{\partial_R \mathcal{L}}{\partial \chi^\ell} \delta \chi^\ell - \sum_\ell \frac{\partial_R \mathcal{L}}{\partial (\partial_\mu \chi^\ell)} \partial_\mu \delta \chi^\ell \\ &= -\delta \mathcal{L}. \end{aligned} \quad (26.7.3)$$

(Here ∂_R denotes the right partial derivative, defined by moving differentiated fermionic variables to the right before differentiating.) Instead, we must define the supersymmetry current as

$$S^\mu \equiv N^\mu + K^\mu, \quad (26.7.4)$$

which Eqs. (26.7.1) and (26.7.3) tell us is conserved:

$$\partial_\mu S^\mu = 0. \quad (26.7.5)$$

The second complication is that the change $\delta\chi^\ell$ in a canonical field χ^ℓ under a supersymmetry transformation is not just a function of canonical fields, but also involves their canonical conjugates. For instance, Eq. (26.3.15) shows that the change in the ψ -component of a chiral scalar superfield involves the time derivative of the ϕ -component. As a result the commutator of the Noether charge $\int d^3x N^0$ with a general canonical field does not give the supersymmetry transformation of that field. Fortunately, this complication is cancelled by the first complication: when the commutators of $\int d^3x K^0$ as well as $\int d^3x N^0$ with the fields are taken into account, the operator $\int d^3x S^0$ does generate supersymmetry transformations¹ in the sense that

$$\left[\int d^3x (\bar{\alpha} S^0), \chi^\ell \right] = i\delta\chi^\ell, \quad (26.7.6)$$

consistently with Eqs. (26.2.1) and (26.2.8).

¹This is a general result for currents constructed in this way. For instance, consider a Lagrangian L (not Lagrangian density) that depends on a set of canonical variables q^n and their time derivatives $\dot{q}^n$, with no constraints of any class. In quantum field theory the label n includes spatial coordinates as well as discrete spin and species labels, and $L = \int d^3x \mathcal{L}$. Our assumption here that the Lagrangian density is invariant up to spacetime derivatives under some infinitesimal transformation δ means that δL is the time derivative of some functional F . That is,

$$\sum_n \frac{\partial L}{\partial q^n} \delta q^n + \sum_n \frac{\partial L}{\partial \dot{q}^n} \delta \dot{q}^n = \frac{d}{dt} F.$$

Using the canonical equations of motion, this can be written as a conservation law $\dot{Q} = 0$, where the conserved charge is

$$Q = - \sum_n \frac{\partial L}{\partial \dot{q}^n} \delta q^n + F.$$

In our case here, $Q = \int d^3x [N^0 + K^0]$. We assume the usual unconstrained commutation relations

$$\left[\frac{\partial L}{\partial \dot{q}^n}, q^m \right] = -i\delta_n^m, \quad [q^n, q^m] = 0,$$

and find the commutator

$$[Q, q^m] = i\delta q^m - \sum_{n\ell} \frac{\partial L}{\partial \dot{q}^\ell} \frac{\partial \delta q^\ell}{\partial \dot{q}^n} [\dot{q}^n, q^m] + \sum_n \frac{\partial F}{\partial \dot{q}^n} [\dot{q}^n, q^m].$$

To evaluate the second and third terms, we note that the second time derivatives $\ddot{q}^n$ appear linearly in the invariance condition, so their coefficients must match: even without using the equations of motion, we have

$$\sum_\ell \frac{\partial L}{\partial \dot{q}^\ell} \frac{\partial \delta q^\ell}{\partial \dot{q}^n} = \frac{\partial F}{\partial \dot{q}^n}.$$

The second and third terms in the commutator therefore cancel, leaving us with the desired result

$$[Q, q^m] = i\delta q^m.$$

Taking the time derivative gives also

$$[Q, \dot{q}^m] = i\delta \dot{q}^m.$$

This result has been extended to theories with constraints [7].

For example, we may derive an explicit formula for the supersymmetry current in the general renormalizable theory of left-chiral superfields Φ_n , which can be used to check that it does generate supersymmetry transformations, in the sense of Eq. (26.7.6). The Lagrangian density (26.4.7) for this theory may be put in the form

$$\mathcal{L} = \sum_n \left[-\partial_\mu \phi_n^* \partial^\mu \phi_n - \frac{1}{2} (\overline{\psi_{Ln}} \gamma^\mu \partial_\mu \psi_{Ln}) - \frac{1}{2} (\overline{\psi_{Rn}} \gamma^\mu \partial_\mu \psi_{Rn}) \right] + \text{non-derivative terms}. \quad (26.7.7)$$

Using the transformation rules (26.3.15), (26.3.17), (26.3.18), and (26.3.20) (with $\tilde{\phi} = \phi^*$), the Noether current defined by Eq. (26.7.2) is

$$N^\mu = \frac{1}{\sqrt{2}} \sum_n \left[2(\partial^\mu \phi_n^*) \psi_{nL} + 2(\partial^\mu \phi_n) \psi_{nR} + (\not{\partial} \phi_n) \gamma^\mu \psi_{nR} + (\not{\partial} \phi_n^*) \gamma^\mu \psi_{nL} \right. \\ \left. - \mathcal{F}_n \gamma^\mu \psi_{nR} - \mathcal{F}_n^* \gamma^\mu \psi_{nL} \right]. \quad (26.7.8)$$

We can calculate the change in the Lagrangian density either directly, or, more easily, by noting that the changes of D -terms and $\mathcal{F}$ -terms under supersymmetry transformations are given respectively by Eqs. (26.2.17) and (26.3.16). Either way, we find that the current K^μ in Eq. (26.7.1) is

$$K^\mu = \frac{1}{\sqrt{2}} \sum_n \gamma^\mu \left[-(\not{\partial} \phi_n) \psi_{nR} - (\not{\partial} \phi_n^*) \psi_{nL} + \mathcal{F}_n^* \psi_{nL} + \mathcal{F}_n \psi_{nR} \right. \\ \left. + 2 \left(\frac{\partial f(\phi)}{\partial \phi_n} \right) \psi_{nL} + 2 \left(\frac{\partial f(\phi)}{\partial \phi_n} \right)^* \psi_{nR} \right]. \quad (26.7.9)$$

Adding (26.7.8) and (26.7.9) gives the supersymmetry current for this class of theories

$$S^\mu = \sqrt{2} \sum_n \left[(\not{\partial} \phi_n) \gamma^\mu \psi_{nR} + (\not{\partial} \phi_n^*) \gamma^\mu \psi_{nL} \right. \\ \left. + \left(\frac{\partial f}{\partial \phi_n} \right) \gamma^\mu \psi_{nL} + \left(\frac{\partial f}{\partial \phi_n} \right)^* \gamma^\mu \psi_{nR} \right]. \quad (26.7.10)$$

It is straightforward then to use the canonical commutation and anticommutation relations to verify that $\int d^3x S^0$ satisfies the commutation relations (26.7.6).

There is another definition of symmetry currents, in terms of the response of the matter action to a local symmetry transformation, which is particularly useful when the associated symmetries are ‘gauged,’ as supersymmetry will be when we turn to supergravity theories in Chapter 31. In the absence of supergravity fields, the action is not invariant under local supersymmetry transformations. If we make such a transformation with a spacetime-dependent parameter $\alpha(x)$, the action will change by an amount that, in order to vanish when $\alpha(x)$ is constant, must (even when the field equations are not satisfied) be of the form

$$\delta I = - \int d^4x ((\partial_\mu \bar{\alpha}(x)) S^\mu(x)), \quad (26.7.11)$$

with $S^\mu(x)$ a four-vector of Majorana spinor operator coefficients. This does not define $S^\mu(x)$ uniquely, because when we generalize global supersymmetry transformations to local

transformations, we might in general give the change $\delta\chi$ of a field χ under a local supersymmetry transformation an arbitrary dependence on the derivatives of $\alpha(x)$. There is, however, one way of defining local supersymmetry transformations that guarantees that the coefficient $S^\mu(x)$ in Eq. (26.7.11) is the same as the current defined by Eq. (26.7.4), which as we have seen generates the symmetry transformations in the sense of Eq. (26.7.6). It is to specify that *derivatives of $\alpha(x)$ do not appear in the supersymmetry transformation of the canonical or auxiliary fields χ^ℓ* . For instance, the local versions of the transformation rules (26.3.15)–(26.3.17) for the components of a left-chiral superfield are

$$\delta\psi_L(x) = \sqrt{2}\partial_\mu\phi(x)\gamma^\mu\alpha_R(x)\phi(x) + \sqrt{2}\mathcal{F}(x)\alpha_L(x), \quad (26.7.12)$$

$$\delta\mathcal{F}(x) = \sqrt{2}(\bar{\alpha}_L(x)\not{\partial}\psi_L(x)), \quad (26.7.13)$$

$$\delta\phi(x) = \sqrt{2}(\bar{\alpha}_R(x)\psi_L(x)). \quad (26.7.14)$$

Eq. (26.3.21) shows that the superfield may be expressed in terms of its component fields at x_+^μ without derivatives, so the transformation rule for the superfield may be expressed as

$$\delta\Phi(x, \theta) = (\bar{\alpha}(x_+)\mathcal{Q})\Phi(x, \theta), \quad (26.7.15)$$

where $\mathcal{Q}$ is the operator (26.2.2).

With local supersymmetry transformations defined in this way, the change of the action that they induce consists of two terms. First, although the variation in canonical fields under a supersymmetry transformation does not involve derivatives of $\alpha(x)$, the variation in derivatives of canonical fields does. This produces a change in the Lagrangian density which is the same as Eq. (26.7.2), except that $\bar{\alpha}$ is replaced with $\partial_\mu\bar{\alpha}$:

$$\delta_1 I = - \int d^4x ([\partial_\mu\bar{\alpha}(x)]N^\mu(x)).$$

The second term in the change of the action arises from the fact that the Lagrangian density is not invariant even under the part of the supersymmetry transformation that does not involve derivatives of $\alpha(x)$. According to Eq. (26.7.1), this produces a change in the action

$$\delta_2 I = \int d^4x (\bar{\alpha}(x)\partial_\mu K^\mu(x)) = - \int d^4x ((\partial_\mu\bar{\alpha}(x))K^\mu(x)).$$

Adding $\delta_1 I$ and $\delta_2 I$ gives a total change in the action of the form (26.7.11), with $S^\mu(x)$ given by Eq. (26.7.4), as was to be shown.

Even with this specification of the transformation property of the component fields, the supersymmetry current $S^\mu(x)$ is not uniquely specified by Eq. (26.7.11), because we could always introduce a modified current

$$S_{\text{new}}^\mu = S^\mu + \partial_\nu A^{\mu\nu}, \quad (26.7.16)$$

where $A^{\mu\nu} = -A^{\nu\mu}$ is an arbitrary antisymmetric tensor of Majorana spinors. The term $\partial_\nu A^{\mu\nu}$ is conserved whether or not the field equations are satisfied, and its time component is a space derivative, so $\int d^3x S_{\text{new}}^0 = \int d^3x S^0$, leaving Eq. (26.7.6) unchanged.

There is in fact a particular choice of $A^{\mu\nu}$ with the convenient feature that $\gamma_\mu S_{\text{new}}^\mu$ turns out to be a measure of the violation of scale invariance by the theory. By using the Dirac equations derived from the Lagrangian density (26.4.7):

$$\not{\partial}\psi_{mL} = -\sum_n \left(\frac{\partial^2 f(\phi)}{\partial\phi_m \partial\phi_n} \right)^* \psi_{nR}, \quad \not{\partial}\psi_{mR} = -\sum_n \left(\frac{\partial^2 f(\phi)}{\partial\phi_m \partial\phi_n} \right) \psi_{nL}, \quad (26.7.17)$$

it is straightforward to calculate that

$$\begin{aligned} \gamma_\mu S^\mu = -2\sqrt{2} \sum_n \bigg\{ & \not{\partial}(\phi_n \psi_{nR} + \phi_n^* \psi_{nL}) \\ & + \left(\sum_m \phi_m \frac{\partial^2 f(\phi)}{\partial\phi_n \partial\phi_m} - 2 \frac{\partial f(\phi)}{\partial\phi_n} \right) \psi_{nL} \\ & + \left(\sum_m \phi_m \frac{\partial^2 f(\phi)}{\partial\phi_n \partial\phi_m} - 2 \frac{\partial f(\phi)}{\partial\phi_n} \right)^* \psi_{nR} \bigg\}. \end{aligned}$$

We can eliminate the first term by introducing a modified supersymmetry current of the general type of Eq. (26.7.16):

$$S_{\text{new}}^\mu = S^\mu + \frac{\sqrt{2}}{3} [\gamma^\mu, \gamma^\nu] \sum_n \partial_\nu (\phi_n \psi_{nR} + \phi_n^* \psi_{nL}), \quad (26.7.18)$$

for which

$$\begin{aligned} \gamma_\mu S_{\text{new}}^\mu = -2\sqrt{2} \sum_n \bigg\{ & \left(\sum_m \phi_m \frac{\partial^2 f(\phi)}{\partial\phi_n \partial\phi_m} - 2 \frac{\partial f(\phi)}{\partial\phi_n} \right) \psi_{nL} \\ & + \left(\sum_m \phi_m \frac{\partial^2 f(\phi)}{\partial\phi_n \partial\phi_m} - 2 \frac{\partial f(\phi)}{\partial\phi_n} \right)^* \psi_{nR} \bigg\}. \end{aligned} \quad (26.7.19)$$

The right-hand side vanishes for a scale-invariant Lagrangian density, with $f(\Phi)$ a homogeneous polynomial of third order in the Φ_n .

We now turn to the supersymmetry transformation property of the supersymmetry current. It is straightforward to check that the current given by Eqs. (26.7.18) and (26.7.10) is related to the ω -component ω_μ^Θ of a real non-chiral superfield Θ_μ by²

$$S_{\text{new}}^\mu = -2\omega^{\Theta\mu} + 2\gamma^\mu \gamma^\nu \omega_\nu^\Theta, \quad (26.7.20)$$

²Here we introduce a notation that will be used extensively in Chapter 31; following Eq. (26.2.10), the components C^S , ω^S , M^S , N^S , V_ν^S , λ^S , and D^S of an arbitrary superfield $S(x, \theta)$ are defined by the expansion

$$\begin{aligned} S(x, \theta) = & C^S(x) - i(\bar{\theta}\gamma_5\omega^S(x)) - \frac{i}{2}(\bar{\theta}\gamma_5\theta)M^S(x) - \frac{1}{2}(\bar{\theta}\theta)N^S(x) \\ & + \frac{i}{2}(\bar{\theta}\gamma_5\gamma^\nu\theta)V_\nu^S(x) - i(\bar{\theta}\gamma_5\theta)\left(\bar{\theta}\left[\lambda^S(x) + \frac{1}{2}\not{\partial}\omega^S(x)\right]\right) \\ & - \frac{1}{4}(\bar{\theta}\gamma_5\theta)^2\left[D^S(x) + \frac{1}{2}\square C^S(x)\right]. \end{aligned}$$

where

$$\Theta_\mu = \frac{i}{12} \sum_n \left[4\Phi_n^* \partial_\mu \Phi_n - 4\Phi_n \partial_\mu \Phi_n^* + ((\bar{\mathcal{D}}\Phi_n^*)\gamma_\mu(\mathcal{D}\Phi_n)) \right]. \quad (26.7.21)$$

The superfield Θ^μ is known as the *supercurrent*.

The supercurrent obeys a conservation law that incorporates the conservation of the supersymmetry current (26.7.20) and much else besides. To derive it, we may use the anticommutation relation (26.2.30) to write³

$$[\mathcal{D}_R, (\bar{\mathcal{D}}_L \mathcal{D}_L)] = -4\not{D}_L.$$

Together with the chirality conditions $\mathcal{D}_R \Phi_n = \mathcal{D}_L \Phi_n^* = 0$, this gives

$$\begin{aligned} \gamma^\mu \mathcal{D}_L \sum_n [\Phi_n^* \partial_\mu \Phi_n - \Phi_n \partial_\mu \Phi_n^*] &= -\frac{1}{4} \sum_n \Phi_n^* \mathcal{D}_R (\bar{\mathcal{D}}_L \mathcal{D}_L) \Phi_n \\ &\quad - \sum_n (\not{D}\Phi_n^*) \mathcal{D}_L \Phi_n \end{aligned}$$

and

$$\gamma^\mu \mathcal{D}_L \sum_n ((\bar{\mathcal{D}}\Phi_n^*)\gamma_\mu(\mathcal{D}\Phi_n)) = 4 \sum_n (\not{D}\Phi_n^*) \mathcal{D}\Phi_n + 2 \sum_n \mathcal{D}\Phi_n^* (\bar{\mathcal{D}}_L \mathcal{D}_L) \Phi_n,$$

so that the superfield (26.7.21) satisfies

$$\begin{aligned} \gamma^\mu \mathcal{D}_L \Theta_\mu &= \frac{i}{6} \sum_n (\mathcal{D}_R \Phi_n^*) (\bar{\mathcal{D}}_L \mathcal{D}_L) \Phi_n \\ &\quad - \frac{i}{12} \sum_n \Phi_n^* \mathcal{D}_R (\bar{\mathcal{D}}_L \mathcal{D}_L) \Phi_n. \end{aligned} \quad (26.7.22)$$

We saw in Section 26.6 that the field equations for the Lagrangian density (26.4.7) may be expressed in the form

$$(\bar{\mathcal{D}}_L \mathcal{D}_L) \Phi_n = -4 \left(\frac{\partial f(\Phi)}{\partial \Phi_n} \right)^*. \quad (26.7.23)$$

Using this in Eq. (26.7.22) gives finally

$$\begin{aligned} \gamma^\mu \mathcal{D}_L \Theta_\mu &= -\frac{2i}{3} \sum_n (\mathcal{D}_R \Phi_n^*) \left(\frac{\partial f}{\partial \Phi_n} \right)^* + \frac{i}{3} \sum_n \Phi_n^* \mathcal{D}_R \left(\frac{\partial f}{\partial \Phi_n} \right)^* \\ &= \frac{i}{3} \mathcal{D}_R \left[\sum_n \Phi_n \frac{\partial f(\Phi)}{\partial \Phi_n} - 3f(\Phi) \right]^*. \end{aligned} \quad (26.7.24)$$

The Hermitian adjoint of Eq. (26.7.24) is

$$\gamma^\mu \mathcal{D}_R \Theta_\mu = -\frac{i}{3} \mathcal{D}_L \left[\sum_n \Phi_n \frac{\partial f(\Phi)}{\partial \Phi_n} - 3f(\Phi) \right]. \quad (26.7.25)$$

³It should be noted that $\bar{\mathcal{D}}_L$ and $\bar{\mathcal{D}}_R$ are the left- and right-handed components of the covariant adjoint $\bar{\mathcal{D}}$, rather than the covariant adjoints $\bar{\mathcal{D}}_L$ and $\bar{\mathcal{D}}_R$ of $\mathcal{D}_L$ and $\mathcal{D}_R$.

The sum of this and Eq. (26.7.24) then gives the conservation law

$$\gamma^\mu \mathcal{D}\Theta_\mu = \mathcal{D}X, \quad (26.7.26)$$

where X is a real chiral superfield given (up to an additive constant) in this class of theories by

$$X = \frac{2}{3} \text{Im} \left[\sum_n \Phi_n \frac{\partial f(\Phi)}{\partial \Phi_n} - 3f(\Phi) \right]. \quad (26.7.27)$$

Although it has been derived here only for renormalizable theories of chiral superfields, we may expect that the conservation law (26.7.26) holds more generally, though of course with X not necessarily given by Eq. (26.7.27), because of the other conservation laws that it incorporates. (A generalized formula for X will be given in Section 31.4.) To derive these relations, we must use Eq. (26.2.10) to express Θ_μ in terms of components C_μ^Θ , ω_μ^Θ , etc. and use Eq. (26.3.9) to express the chiral superfield X in terms of components A^X , ψ^X , etc. With the aid of Eqs. (26.A.9), (26.A.16), (26.A.17), and the Dirac matrix identities

$$[\gamma^\rho, \gamma^\sigma] = -\frac{i}{2} \epsilon^{\rho\sigma\mu\nu} \gamma_5 [\gamma_\mu, \gamma_\nu], \quad (26.7.28)$$

$$\gamma^\mu \gamma^\rho \gamma^\nu = \eta^{\mu\rho} \gamma^\nu - \eta^{\mu\nu} \gamma^\rho + \eta^{\nu\rho} \gamma^\mu + i\gamma_5 \epsilon^{\mu\nu\rho\sigma} \gamma_\sigma, \quad (26.7.29)$$

we can then expand both sides of Eq. (26.7.26) in the terms

$$\begin{aligned} &1, \quad \theta, \quad \gamma_5 \theta, \quad \gamma^\nu \theta, \quad \gamma_5 \gamma^\nu \theta, \quad \gamma_5 [\gamma^\mu, \gamma^\nu] \theta, \\ &(\bar{\theta}\theta), \quad (\bar{\theta}\gamma_5 \theta), \quad (\bar{\theta}\gamma_5 \gamma^\nu \theta), \\ &\theta(\bar{\theta}\gamma_5 \theta), \quad \gamma_5 \theta(\bar{\theta}\gamma_5 \theta), \quad \gamma^\nu \theta(\bar{\theta}\gamma_5 \theta), \\ &\gamma^\nu \gamma_5 \theta(\bar{\theta}\gamma_5 \theta), \quad [\gamma^\rho, \gamma^\sigma] \theta(\bar{\theta}\gamma_5 \theta), \quad (\bar{\theta}\gamma_5 \theta)^2. \end{aligned}$$

Matching the coefficients of $1, \theta, \gamma_5 \theta, \gamma^\nu \theta, \gamma_5 \gamma^\nu \theta, \gamma_5 [\gamma^\mu, \gamma^\nu] \theta$, respectively, yields the results⁴

$$\psi^X = -i\gamma_5 \gamma^\mu \omega_\mu^\Theta, \quad (26.7.30)$$

$$F^X = \partial^\mu C_\mu^\Theta, \quad (26.7.31)$$

$$G^X = (V^\Theta)^\mu{}_\mu, \quad (26.7.32)$$

$$\partial_\mu A^X = -N_\mu^\Theta, \quad (26.7.33)$$

$$\partial_\mu B^X = M_\mu^\Theta, \quad (26.7.34)$$

$$0 = V_{\mu\nu}^\Theta - V_{\nu\mu}^\Theta + \epsilon_{\mu\nu\rho\sigma} \partial^\sigma C^{\Theta\rho}. \quad (26.7.35)$$

Matching the coefficients of either $(\bar{\theta}\theta)$ or $(\bar{\theta}\gamma_5 \theta)$ yields the same result:

$$0 = \gamma^\mu \lambda_\mu^\Theta, \quad (26.7.36)$$

and matching the coefficients of $(\bar{\theta}\gamma_5 \gamma^\nu \theta)$ yields the result

$$-i\gamma_5 [\gamma^\nu, \not{\partial}] \psi^X = 2\gamma^\mu \gamma^\nu \lambda_\mu^\Theta + \gamma^\mu [\gamma^\nu, \not{\partial}] \omega_\mu^\Theta. \quad (26.7.37)$$

⁴Note that $V_{\mu\nu}^\Theta$ is the V_ν -component of Θ_μ , not the V_μ -component of Θ_ν .

From Eqs. (26.7.30), (26.7.36), and (26.7.37) we obtain the conservation of the supersymmetry current (26.7.20):

$$0 = \partial_\mu S_{\text{new}}^\mu = -2\partial^\mu \omega_\mu^\Theta + 2\bar{\theta}\gamma^\mu \omega_\mu^\Theta, \quad (26.7.38)$$

and a relation among λ_μ^Θ and ω_μ^Θ :

$$\lambda_\nu^\Theta = -\bar{\theta}\omega_\nu^\Theta + \partial_\nu \gamma^\mu \omega_\mu^\Theta. \quad (26.7.39)$$

Matching the coefficients of $\theta(\bar{\theta}\gamma_5\theta)$ and $\gamma_5\theta(\bar{\theta}\gamma_5\theta)$ gives relations that can be obtained by taking the divergence of Eqs. (26.7.34) and (26.7.33), respectively. Matching the coefficients of $\gamma^\rho\theta(\bar{\theta}\gamma_5\theta)$ gives

$$\partial_\rho G^X = \partial^\mu V_{\mu\rho}^\Theta + \partial^\mu V_{\rho\mu}^\Theta - \partial_\rho V^{\Theta\lambda}{}_\lambda, \quad (26.7.40)$$

which, combined with Eq. (26.7.32), yields the conservation law

$$\partial_\mu T^{\mu\nu} = 0, \quad (26.7.41)$$

where $T^{\mu\nu}$ is the symmetric tensor

$$T_{\mu\nu} \equiv -\frac{1}{2}V_{\mu\nu}^\Theta - \frac{1}{2}V_{\nu\mu}^\Theta + \eta_{\mu\nu}V^{\Theta\lambda}{}_\lambda. \quad (26.7.42)$$

Matching the coefficients of $\gamma^\rho\gamma_5\theta(\bar{\theta}\gamma_5\theta)$ gives

$$\partial_\mu F^X = 2D_\mu^\Theta + \square C_\mu^\Theta + \epsilon_{\rho\nu\sigma\mu}\partial^\nu V^{\Theta\rho\sigma}, \quad (26.7.43)$$

which, with Eqs. (26.7.31) and (27.6.35), yields a relation between D_μ^Θ and C_μ^Θ :

$$D_\mu^\Theta = -\square C_\mu^\Theta + \partial_\mu \partial^\nu C_\nu^\Theta. \quad (26.7.44)$$

Matching the coefficients of $[\gamma^\rho, \gamma^\sigma]\theta(\bar{\theta}\gamma_5\theta)$ and $(\bar{\theta}\gamma_5\theta)^2$ gives results that already follow from Eq. (26.7.34) and from Eqs. (26.7.38) and (26.7.39), respectively.

The conserved symmetric tensor $T^{\mu\nu}$ may be identified as the energy-momentum tensor of the system. To check this, we use Eqs. (26.1.18) and (26.2.12) to write the change in $\omega_\mu^\Theta(x)$ under a supersymmetry transformation with infinitesimal parameter α as

$$\begin{aligned} \delta\omega_\mu^\Theta &= -i[(\bar{Q}\alpha), \omega_\mu^\Theta] = +i[\omega_\mu^\Theta, (\bar{Q}\alpha)] \\ &= (-i\gamma_5\bar{\theta}C_\mu^\Theta - M_\mu^\Theta + i\gamma_5N_\mu^\Theta + \gamma^\nu V_{\mu\nu}^\Theta)\alpha. \end{aligned}$$

Eqs. (26.7.33)–(26.7.35) allow us to put this in the form

$$\begin{aligned} i\{\omega_\mu^\Theta, \bar{Q}\} &= \frac{1}{2}\gamma^\nu(V_{\mu\nu}^\Theta + V_{\nu\mu}^\Theta) - \partial_\mu(B^X + \gamma_5A^X) - i\gamma_5\bar{\theta}C_\mu^\Theta \\ &\quad + \frac{1}{2}\epsilon_{\mu\nu\kappa\sigma}\gamma^\nu\partial^\kappa C^{\Theta\sigma}. \end{aligned}$$

In terms of the currents (26.7.20) and (26.7.42), this reads

$$\begin{aligned} i\{S_{\text{new}}^\mu, \bar{Q}\} &= 2\gamma_\nu T^{\mu\nu} + 2(\partial^\mu - \gamma^\mu\bar{\theta})(B^X + \gamma_5A^X) - \epsilon^{\mu\nu\kappa\sigma}\gamma_\nu\partial_\kappa C_\sigma^\Theta \\ &\quad + 2i\gamma_5\left(\bar{\theta}C^{\Theta\mu} - \gamma^\mu\gamma^\lambda\bar{\theta}C_\lambda^\Theta - \frac{1}{2}\gamma^\mu[\bar{\theta}, \gamma^\sigma]C_\sigma^\Theta\right). \end{aligned} \quad (26.7.45)$$

For $\mu = 0$ all the terms on the right except the first are space derivatives, and therefore vanish when we integrate over space, leaving us with

$$i \left\{ \int d^3x S_{\text{new}}^0, \bar{Q} \right\} = 2\gamma_\nu \int d^3x T^{0\nu}. \quad (26.7.46)$$

We have defined the supersymmetry current S_{new}^μ to give $\int d^3x S_{\text{new}}^0 = Q$, so the fundamental anticommutation relation (25.2.36) tells us that

$$\int d^3x T^{0\nu} = P^\nu, \quad (26.7.47)$$

which, with the conservation condition (26.7.41), allows us to identify $T^{\mu\nu}$ as the energy-momentum tensor.

It is important to note *which* energy-momentum tensor we have constructed in this way. Either directly from Eq. (26.7.21), or by considering the supersymmetry transformation of the current (26.7.18), we can calculate that the energy-momentum tensor $T^{\mu\nu}$ for renormalizable theories of chiral superfields is

$$\begin{aligned} T^{\mu\nu} = & \sum_n [\partial^\mu \phi_n^* \partial^\nu \phi_n + \partial^\nu \phi_n^* \partial^\mu \phi_n] \\ & - \eta^{\mu\nu} \sum_n \left[\partial^\lambda \phi_n^* \partial_\lambda \phi_n + \left| \frac{\partial f(\phi)}{\partial \phi_n} \right|^2 \right] \\ & + \frac{1}{3} (\eta^{\mu\nu} \square - \partial^\mu \partial^\nu) \sum_n |\phi_n|^2 + \dots, \end{aligned} \quad (26.7.48)$$

where the dots denote terms involving fermions, which do not concern us here. We see that the last term, which is related by supersymmetry to the correction term in Eq. (26.7.18), has the effect that the energy-momentum tensor is traceless for the massless free-field theory with no superpotential, in which case $\square \phi_n = 0$. A simple calculation shows that $T^{\mu\nu}$ is also traceless more generally for scale-invariant theories, with $f(\phi)$ a homogeneous polynomial of third order in the ϕ_n .

Supersymmetry also imposes an interesting relation among violations of scale invariance and R conservation. Eqs. (26.7.30)–(26.7.32) show that $\gamma_\mu S_{\text{new}}^\mu = 6\gamma_\mu \omega^{\Theta\mu}$, $\partial^\mu C_\mu^\Theta$, and $T^\lambda{}_\lambda = 2V^{\Theta\mu}{}_\mu$ (which measures the violation of scale invariance) are proportional to components of a chiral superfield X , so if any one of these vanishes as an operator equation (that is, not just for some particular field configuration), then they all do. In this case, we can show that $C^{\Theta\rho}$ is proportional to the current of an R quantum number. To see this, note that Eq. (26.2.11) gives

$$\delta C_\sigma^\Theta = i[C_\sigma^\Theta, (\bar{\alpha}Q)] = i(\bar{\alpha}\gamma_5\omega_\sigma^\Theta),$$

so that in general

$$[C_\sigma^\Theta, Q] = \gamma_5\omega_\sigma^\Theta. \quad (26.7.49)$$

We have seen that if C_σ^Θ is conserved then $\gamma_\mu S^\mu = 0$, so that Eq. (26.7.20) gives $S_\sigma = -2\omega_\sigma^\Theta$. Setting $\sigma = 0$ and integrating over $\mathbf{x}$ in Eq. (26.7.49) then gives

$$\left[\int d^3x C^{0\Theta}, Q \right] = -\frac{1}{2}\gamma_5 Q. \quad (26.7.50)$$

We can therefore introduce a current

$$\mathcal{R}^\mu \equiv 2C^{\mu\Theta}, \quad (26.7.51)$$

which *if* conserved is the current of a quantum number $\mathcal{R} \equiv \int d^3x \mathcal{R}^0$ for which Q_L and Q_R destroy the values $+1$ and -1 , respectively. Since the commutator of Q_L with a scalar superfield Φ involves a term $\partial\Phi/\partial\theta_L$, this means that θ_L carries the $\mathcal{R}$ value $+1$, in accord with the usual definition. A theory in which the superfield X vanishes, or, equivalently, in which $T^\mu{}_\mu$, $\gamma_\mu S^\mu$, and $\partial_\mu \mathcal{R}^\mu$ all vanish, is invariant under an enlarged set of supersymmetry transformations, generated by the superconformal algebra described at the end of Section 25.2.

In scale-invariant theories the value of the $\mathcal{R}$ quantum number carried by various superfields is fixed by the structure of the Lagrangian. For instance, in a scale-invariant theory of chiral scalar superfields, the superpotential must be a homogeneous polynomial of third order in the superfields. The $\mathcal{F}$ -term of the superpotential is proportional to the coefficient of θ_L^2 , which has $\mathcal{R}$ quantum number $+2$, so the $\mathcal{R}$ quantum number of the $\mathcal{F}$ -term of the superpotential is the $\mathcal{R}$ quantum number of the superpotential itself minus two. $\mathcal{R}$ invariance then requires that we give the scalar superfields an $\mathcal{R}$ quantum number $+2/3$, so that the superpotential will have $\mathcal{R}$ quantum number $+2$, and its $\mathcal{F}$ -term will have $\mathcal{R}$ quantum number zero. That is, the scalar components ϕ_n have $\mathcal{R} = 2/3$ and the spinor components ψ_{nL} (proportional to the coefficient of θ_L in the superfield) have $\mathcal{R} = -1/3$. This can be verified by calculating the current $\mathcal{R}^\mu$ from the C -term of the supercurrent (26.7.21) for this class of theories:

$$\mathcal{R}_\mu = \frac{2i}{3}[\phi^* \partial_\mu \phi - \phi \partial_\mu \phi^*] - \frac{i}{6}(\bar{\psi} \gamma_\mu \gamma_5 \psi). \quad (26.7.52)$$

(The second term contains an extra factor of $1/2$ because ψ is a Majorana spinor.)

Quantum corrections can introduce violations of $\mathcal{R}$ invariance (through Adler–Bell–Jackiw anomalies) and of scale invariance (through the renormalization group running of the coupling constants) but supersymmetry continues to impose a relation between these symmetry violations despite these corrections [7a]. We will see an example of this in Section 29.3.

* * *

The conservation condition (26.7.26) does not uniquely determine either the supercurrent Θ^μ or the associated chiral superfield X . In particular, we may add to Θ^μ a change

$$\Delta\Theta^\mu = \partial^\mu Y, \quad (26.7.53)$$

with Y an arbitrary chiral superfield. The left-hand side of Eq. (26.7.26) is then changed by

$$\gamma_\mu \mathcal{D} \Delta\Theta^\mu = \not{D} Y.$$

For a left-chiral superfield Y_L the chirality condition $\mathcal{D}_R Y_L = 0$ and the anticommutation relations (26.2.30) give

$$\begin{aligned}\not{D}\mathcal{D}_\alpha Y &= -\frac{1}{2} [\{\mathcal{D}_L, \bar{\mathcal{D}}_R\}\mathcal{D}_R]_\alpha Y_L \\ &= -\frac{1}{2} \left[\mathcal{D}_{L\alpha}(\bar{\mathcal{D}}_R \mathcal{D}_R) Y_L + \sum_\beta \bar{\mathcal{D}}_{L\beta} \mathcal{D}_{R\alpha} \mathcal{D}_{L\beta} Y_L \right].\end{aligned}$$

The matrix $\epsilon\gamma_5$ in $\bar{\mathcal{D}}_{L\beta}$ in the second term on the right of the above expression may be moved to the final operator $\mathcal{D}_{L\beta}$, so that the chirality condition and anticommutation relations give

$$\sum_\beta \bar{\mathcal{D}}_{L\beta} \mathcal{D}_{R\alpha} \mathcal{D}_{L\beta} Y_L = - \sum_\beta \mathcal{D}_{L\beta} \mathcal{D}_{R\alpha} \bar{\mathcal{D}}_{L\beta} Y_L = 2(\not{D}\mathcal{D})_\alpha Y_L,$$

and hence

$$\not{D}\mathcal{D}Y_L = -\frac{1}{2} [\mathcal{D}(\bar{\mathcal{D}}\mathcal{D})Y_L + 2\not{D}\mathcal{D}Y_L] = -\frac{1}{4}\mathcal{D}(\bar{\mathcal{D}}\mathcal{D})Y_L.$$

The same result can be derived in the same way for any right-chiral superfield, and hence it holds also for an arbitrary sum Y of a left-chiral and a right-chiral superfield

$$\gamma_\mu \mathcal{D}\Delta\Theta^\mu = \not{D}\mathcal{D}Y = -\frac{1}{4}\mathcal{D}(\bar{\mathcal{D}}\mathcal{D})Y. \quad (26.7.54)$$

This is of the same form as the conservation condition (26.7.26), with the associated chiral superfield X changed by the chiral superfield

$$\Delta X = -\frac{1}{4}(\bar{\mathcal{D}}\mathcal{D})Y. \quad (26.7.55)$$

It is easy to check that the addition of $\Delta\Theta^\mu$ to Θ^μ changes $T^{\mu 0}$ and S_{new}^0 only by space derivatives, and therefore does not change the energy-momentum four-vector P^μ or the supercharge Q .

We saw in Section 26.6 that any chiral superfield X may be expressed in the form $X = (\bar{\mathcal{D}}\mathcal{D})S$, and hence can be removed by adding a term of the form (26.7.55) to Θ^μ with $Y = 4S$. But in general S and the new Θ^μ constructed in this way will not be local. This situation is already familiar from our experience with triangle anomalies, discussed in Chapter 22—we saw there that, although it is always possible to construct terms that if added to the Lagrangian density would cancel these anomalies, in general these terms would not be local, and hence must be excluded from the Lagrangian density. There are chiral superfields that can be expressed as $(\bar{\mathcal{D}}\mathcal{D})S$ with S local, and that therefore, if present in the associated chiral superfield X , could be eliminated by adding local terms of the form (26.7.53) to Θ^μ . They include, for instance, a term of the form $\text{Re}(k \partial f(\Phi)/\partial\Phi)$, with k an arbitrary complex constant, because the field equations (26.6.15) and (26.6.16) show that $(\bar{\mathcal{D}}\mathcal{D}) \text{Re}(k^*\Phi) = 4 \text{Re}(k \partial f(\Phi)/\partial\Phi)$. But in general the changes in X that can be made in this way are quite limited.